

991 + 981 , MY 2012 =>

Diagnosis information - Rear-end electronics

00C058.....	1
800708.....	2
800808.....	3
80140B.....	4
80140E.....	5
80150B.....	6
802001.....	7
802101.....	8
802201.....	9
802401.....	10
803208.....	11
805108.....	12
805208.....	13
806005.....	14
806108.....	15
80700E.....	16
80710E.....	17
80720B.....	18
80720E.....	19
81010E.....	20
81010E.....	21
81020E.....	22
81020E.....	23
81030E.....	24
81030E.....	25
81040E.....	26
81040E.....	27
81050E.....	28
81050E.....	29
81060E.....	30
81060E.....	31
81090E.....	32
81090E.....	33
810A0E.....	34
810A0E.....	35
820108.....	36
82020E.....	37
830101.....	38
830102.....	39

83020E.....	40
84010E.....	41
84030E.....	42
84040E.....	43
840501.....	44
842001.....	45
842101.....	46
842201.....	47
842301.....	48
842501.....	49
85020E.....	50
85030E.....	51
85040E.....	52
86010E.....	53
86020E.....	54
87010E.....	55
87020E.....	56
87030E.....	57
87040E.....	58
87060E.....	59
88010E.....	60
88020E.....	61
88030E.....	62
88040E.....	63
88050E.....	64
88060E.....	65
88061E.....	66
88070E.....	67
88080E.....	68
88090E.....	69
880A0E.....	70
880B0E.....	71
880C0E.....	72
880D0E.....	73
880E0E.....	74
880F0E.....	75
880F1E.....	76
89010E.....	77
8A0301.....	78

8B0106.....	79
8B0107.....	81
8B010B.....	83
8C0104.....	85
8C0208.....	86
8C0304.....	87
8C0408.....	88
8C0508.....	89
8C0608.....	90
8C0708.....	91
8C1005.....	92
8C110E.....	93
8C200B.....	94
8C210B.....	95
8C220C.....	96
8C230C.....	97
8C240C.....	98
8C2502.....	99
8C260E.....	100
8C270E.....	101
8D010E.....	102
8D020E.....	103
8E0008.....	104
8E0108.....	105
8E0204.....	106
8E0404.....	107
8E0504.....	108
8E0608.....	109
C11000.....	110
C11001.....	111
C11004.....	112
C11005.....	113
C11006.....	114
C11007.....	115
C13006.....	116
C13010.....	117
C13012.....	118
C13013.....	119
C13014.....	120

C13026.....	121
C13032.....	122
C13035.....	123
C13050.....	124
C13106.....	125
C13113.....	126
C13114.....	127
C13126.....	128
C13135.....	129
C13150.....	130
C13206.....	131
C13210.....	132
C13212.....	133
C13213.....	134
C13214.....	135
C13226.....	136
C13232.....	137
C13235.....	138
C13250.....	139
C13306.....	140
C13310.....	141
C13312.....	142
C13313.....	143
C13314.....	144
C13332.....	145
C13335.....	146
C13350.....	147
C13412.....	148
C13A00.....	149
C13B00.....	150

Diagnostic information - Rear-end electronics

Production mode active/Coding Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- One or more production modes or production interventions are activated.
- No/incorrect coding
- **Rear-end electronics** control unit faulty

Fault effects

- ♦ Comfort functions: Function not available or available only to a limited extent

800708

Diagnosis information - Rear-end electronics

800708 PSM rotation speed signal implausible

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Vehicle speed** > 10 km/h
- **Supply voltage** 7 ... 18 V

Fault setting condition(s)

- Signal **Speed sensor** via CAN > 10 s low

Possible fault causes

- ♦ Fault in system PSM
- Perform diagnosis/fault finding in the following system: PSM
- ♦ CAN data bus Faulty

Fault effects

- ♦ **Electronic steering column lock (ELV)** No function

INFORMATION

Electronic steering column lock (ELV) is not locked even when the fault becomes passive again. The fault must be deleted from the fault memory so that the **Electronic steering column lock (ELV)** can lock again.

Diagnosis information - Rear-end electronics

Stop light switch

Diagnostic conditions

♦ The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- Position **Stop light switch** implausible

80140B

Diagnosis information - Component(s)

Relay, heated rear window

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Short circuit between the lines
- Faulty

Fault effects

The affected function no longer reacts

Diagnosis information - Component(s)

Relay, heated rear window

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Short circuit between the lines
- Faulty

Fault effects

The affected function no longer reacts

Diagnosis information - Rear-end electronics

Rear-end electronics power supply

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Short circuit between the lines
- Faulty

Fault effects

The affected function no longer reacts

Diagnostic information - Rear-end electronics

Lid button, rear / Front lid button

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- The button is stuck or permanently actuated.
- Short circuit between the lines

Fault effects

The affected function no longer reacts

Diagnosis information - Component(s)

Microswitch Luggage compartment Front lid emergency release handle

INFORMATION

Only for vehicles with USA and Canada country specification

INFORMATION

The open circuit fault type is not monitored

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ Handle too long **actuated**

(actuated/held longer than 20 seconds)

- ♦ Line connection between control unit and micro switch faulty **Short circuit to ground** (> 20 seconds)
- ♦ Emergency release handle micro switch is stuck or faulty
- ♦ Rear-end electronics control unit faulty

Fault effects

The affected function no longer reacts

Diagnosis information - Component(s)

Button

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

The button is stuck or permanently actuated.

Fault effects

The affected function no longer reacts

Diagnostic information - Rear-end electronics

Lid button, rear / Front lid button

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- The button is stuck or permanently actuated.
- Short circuit between the lines

Fault effects

The affected function no longer reacts

Diagnosis information - Rear-end electronics

Actuator Rear lid release

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to B+
- Open circuit
- Short circuit between the lines
- Faulty

Fault effects

- Limited function

Diagnosis information - Rear-end electronics

NOTE

For vehicles with **Porsche Vehicle Tracking System (PVTS)**, see **Tequipment (TEQ) instructions**, main group **9063**!

► This fault code relates to the **Porsche Vehicle Tracking System (PVTS)**.
More information is available in attached documents or from the respective service provider.

Diagnosis information - Rear-end electronics

Rear-end electronics Immobilizer

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Control module Rear-end electronics** Immobilizer Authentication data incorrect
- **Control module Rear-end electronics** incomplete taught
- **Control module Rear-end electronics** not taught (**Control module Rear-end electronics** new)
- Different data **DME control module** <-> **Rear-end electronics** during teaching

► Teach >component protection< using >PIWIS Tester II< ⇒ >Diagnostics< ⇒ >Online teaching functions<.

Fault effects

- ♦ Alarm system triggers

Diagnosis information - Rear-end electronics

Rear-end electronics Immobilizer

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- **Control module Rear-end electronics** Immobilizer Authentication data incorrect
- **Control module Rear-end electronics** incomplete taught
- **Control module Rear-end electronics** not taught (**Control module Rear-end electronics** new)
- Different data **DME control module** <-> **Rear-end electronics** during teaching

► Teach >component protection< using >PIWIS Tester II< ⇒ >Diagnostics< ⇒ >Online teaching functions<.

Fault effects

- ♦ Alarm system triggers

Diagnosis information - Rear-end electronics

Voltage converter, Output implausible

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ **Supply line** for the **Rear-end electronics** to the **Voltage converter**: Open circuit/short circuit to ground
- ♦ **Voltage converter** Faulty

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Diagnostic information - Rear-end electronics

Filler flap Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- ♦ **Filler flap** No function

Diagnosis information - Component(s)

Microswitch Luggage compartment Front lid emergency release handle

INFORMATION

Only for vehicles with USA and Canada country specification

INFORMATION

The open circuit fault type is not monitored

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- ♦ Handle too long **actuated**

(actuated/held longer than 20 seconds)

- ♦ Line connection between control unit and micro switch faulty **Short circuit to ground** (> 20 seconds)
- ♦ Emergency release handle micro switch is stuck or faulty
- ♦ Rear-end electronics control unit faulty

Fault effects

The affected function no longer reacts

Diagnosis information - Rear-end electronics

Actuator Rear lid release

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to B+
- Open circuit
- Short circuit between the lines
- Faulty

Fault effects

- Limited function

Diagnosis information - Rear-end electronics

Actuator Rear lid release

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to B+
- Open circuit
- Short circuit between the lines
- Faulty

Fault effects

- Limited function

81010E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender



INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

81020E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender

INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

81030E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender



INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

81040E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender



INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

81050E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender

INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

81060E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender



INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

81090E

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender

INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

Diagnosis information - Component(s)

Spoiler drive Speed sensor

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

Short circuit to ground

Fault effects

The affected function no longer reacts

Diagnostic information - component(s)

Spoiler drive speed sensor
Spoiler drive hall sender



INFORMATION

The position sensor is a hall sender . The extended end position is detected by a hall sensor.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- Short circuit to B+
- Open circuit
- Short circuit between lines
- Fault

Fault effects

Spoiler is de-standardized and moves to a safe position.

Diagnosis information - Component(s)

LIN user

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

No signal/communication

Fault effects

The affected function no longer reacts